# QC/ML prediction of metabolite reaction $\Delta_r G'^{\circ}$

## What works, what doesn't, and where QC actually helps

ModelSEED thermodynamics — discussion notes

August 12, 2026

# The question, and the honest one-line answer

**Goal:** compute standard transformed reaction Gibbs energies $\Delta_r G'^\circ$ for metabolic reactions accurately enough to (a) adjudicate dGPredictor vs. experiment, and (b) cover ModelSEED at large ($\sim$48k reactions).

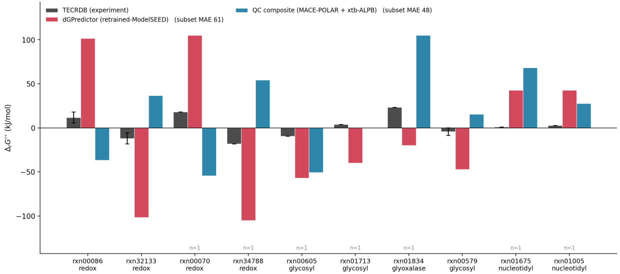
**Method under test:** a physics-only composite

$$G_{aq} = \underbrace{E_{\mathsf{MLIP}}(\text{gas})}_{\text{MACE-POLAR-1}} + \underbrace{\Delta G_{\mathsf{solv}}(\text{ALPB})}_{\text{xtb GFN2}} + \underbrace{G_{\mathsf{RRHO}}}_{\text{xtb}} \rightarrow \text{Alberty transform to pH 7}$$

## Bottom line

On measured chemistry the data-driven methods already win (eQuilibrator 3.0 kJ/mol); the physics composite is at 36 kJ/mol and *cannot be calibrated into competitiveness*. But QC has three narrow, defensible niches — and the failure is a specific, quantified one.

# Where we started: the 10 top-disagreement reactions

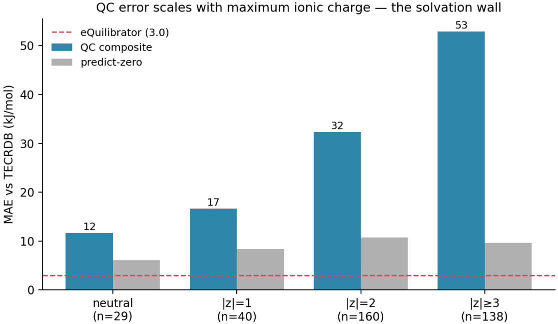



Cherry-picked disagreement subset (NOT representative). Unbiased full-367 MAE: raw QC 36, retrained-dGP 5.7, standard-dGP 3.0, eQuilibrator 3.0. error bars = TECRDB sd; n=1 = single measurement.

| method | MAE | signs |
|---|---|---|
| eQuilibrator | 3.6 | — |
| predict-zero | 10.5 | — |
| **QC + ext. refs** | **16.1** | **10/10** |
| QC composite | 31.7 | 7/10 |
| dGPredictor | 61.2 | 8/10 |

Parameter-free external anchors (redox $E^{\circ\prime}$, isodesmic vs. a measured pyrophosphorylase / sucrose phosphorylase) fix *direction* 10/10 — the adjudication-relevant property. **Still loses to predict-zero on MAE.**

# Full TECRDB run: 367 reactions (80% of all distinct reactions)




- QC MAE 36.1 vs predict-zero 9.7 vs eQuilibrator 3.0
- Error is **monotone in ionic charge** ⇒ it is *solvation*, not the electronic method
- **No stratum beats predict-zero** — not even neutral small molecules (11.6 vs 6.1)

453 compounds, 3 263 conformers. CPU 21 min, GPU 2 min on 80 cores / 8 GPUs.

# The wall is phosphate solvation — ruled out from every direction

| lever tested | result | verdict |
|---|---|---|
| electronic method (r2SCAN-3c vs MLIP) | 14.9 $\rightarrow$ 17.3 | not the cause |
| MLIP choice (UMA vs MACE-POLAR) | $r = 0.98$ | not the cause |
| geometry method (g-xTB vs GFN2 vs DFT) | net $<$1 kJ (terms cancel) | not the cause |
| conformer sampling | error $\perp$ size | not the cause |
| $Mg^{2+}$ speciation | $+$0.3 kJ net | not the cause |
| CPCM-X solvation | phosphate $-150$ kJ | worse |
| explicit-water clusters | over-stabilise polyanions | worse |
| explicit-solvent FEP | GPU-months | infeasible |

### Best achievable phosphate solvation, methods in hand

phenol 8.1 thiol 17.9 carboxylate 15.9 **phosphate 23.9 (ALPB, least bad)**

Phosphate is the most frequent charged group and no available continuum or cheap-cluster method gets it under $\sim$24 kJ/mol.

# The one replicated physical result: the phosphoanhydride bias

- Grouping by number of **P–O–P bonds cleaved**:

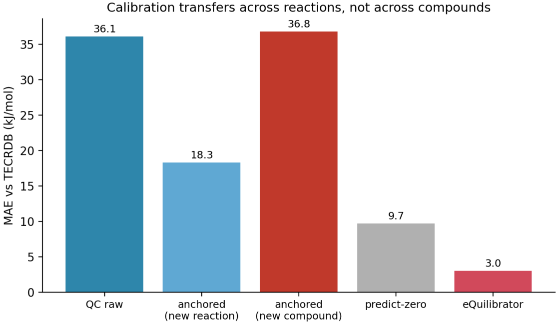
  $\Delta$(P–O–P) $= -1$ : mean signed error $+63.8$ kJ

- Pre-registered $+64 \pm 15$; replicated at $n{=}35$ (was 13), all same sign
- **Not geometry** (g-xTB test: net $<1$ kJ), **not electronic** — it is the solvation of charge *separation* when one polyphosphate becomes two anions

| $\Delta$(P–O–P) | $n$ | mean signed |
|---|---|---|
| $-2$ | 1 | $+169.7$ |
| $-1$ | 35 | $+63.8$ |
| $0$ | 78 | $+12.0$ |

A specific, quantified statement of *where* continuum solvation fails. This is a publishable negative with an identified mechanism.

# Does calibration rescue it? Across reactions yes — across compounds no




- **New reaction, known species** (leave-reaction-out): $36.1 \rightarrow 18.3$
- **New compound** (leave-compound-out): $36.1 \rightarrow 36.8$ — *no benefit*
- Decisive test: fitting the empirical model *directly* to $\Delta_r G'^{\circ}$ gives 7.8; correcting QC gives 23.8; adding QC as a feature moves 7.8→7.6 (∼0.2 kJ)

Per-species anchoring is a lookup table over measured species, not a model. The ModelSEED goal is exactly the *unmeasured* compounds.

# Data-driven baselines (full TECRDB, $n = 364$)

| subset | QC | eQuilibrator | dGP standard | dGP retrained |
|---|---|---|---|---|
| ALL | 35.8 | 3.0 | 3.0 | 5.7 |
| neutral $\lvert z\rvert=0$ | 11.6 | 0.6 | 0.7 | 0.7 |
| $\lvert z\rvert=2$ | 32.4 | 2.9 | 3.0 | 5.5 |
| $\lvert z\rvert\geq 3$ | 50.5 | 4.0 | 3.7 | 7.9 |

- TECRDB is the trained methods' home turf. QC adds nothing here.
- Retrained dGPredictor (ModelSEED features, +coverage) pays the accuracy price only on multiply-charged chemistry: $\lvert z\rvert\geq 3$ goes $3.7 \rightarrow 7.9$.
- Metals: bare-ion spin states silently wrong by 200–470 kJ; now refused loudly. $\sim$1% of reactions, treat as out of scope.

# Where QC *does* have an edge — the forward proposal

Every viable route obeys one rule: never compute an absolute charged-species free energy.

1. **Redox in potential space** (computational H-electrode). Redox looks bad here (41.6) only because absolute-$\Delta_r G'^\circ$ scores the charged cofactor; in $E^{\circ\prime}$ space the backbone cancels. Literature: 2–4 kJ. **Testable now on our 96 NAD(P) reactions.**
2. **QC for speciation** — pKa / tautomer / anomer / stereochemistry, fed into a data-driven $\Delta_r G'^\circ$. This is what group methods are blind to; a single deprotonation cancels the solvation error.
3. **Isodesmic vs. computed analogs** — coverage was capped at 30% only because references were *measured*; compute them and coverage $\rightarrow$ 100%, solvation cancels for charge-balanced residuals.
4. **Cluster-continuum on the $\sim$15 recurring anions**, cached once — the only route that could rescue absolute $\Delta_r G'^\circ$, if phosphate drops below 10.

# Discussion points for tomorrow

- The honest framing: **not one QC problem**. Data-driven methods own measured chemistry (3–6 kJ); QC's role is the narrow complement — novel structures, stereochemistry, speciation, redox potentials.
- Do we invest in **Route 1 (redox potential space)** as the fastest defensible win? (Reuses existing energies, $\sim$1 h.)
- Is a **QC-anchored speciation layer** feeding eQuilibrator worth building? It targets exactly where component-contribution is weak.
- The **ModelSEED-wide run** (14 345 compounds staged) — worth launching, or does the compound-generalization result argue against spending it until the wall falls?
- Is the **phosphoanhydride + solvation-wall** result a paper on its own (QC method limits for polyanion biochemistry)?

# Summary

Physics can't out-predict the data on measured metabolism,

but it tells us *exactly why* — phosphate charge-separation solvation —

and it is uniquely positioned for redox, speciation, and novel stereochemistry,

the places data can't reach.